



INTERNSHIELD

LEARN • PRACTICE • DEFEND • SECURE • INNOVATE

CLOUD COMPUTING & SECURITY

LABS OVERVIEW

FOUR PRACTICAL LABS. ONE SECURE JOURNEY.

A comprehensive, hands-on lab journey to explore cloud security through AWS services, identity management, container security, and vulnerability assessments.



IAM
Identity &
Access Management

S3
Scalable Storage
in the Cloud

KMS
Key Management
Service



AUTHORIZED TRAINING ENVIRONMENT

All labs must be performed only in authorized lab environments using approved tools and targets provided for educational purposes.



PRACTICAL
Hands-on
learning



SECURE
Security-first
approach



REAL-WORLD
Industry
relevant



SKILL BUILDER
Stronger skills
for tomorrow

LAB 01



PERFORMING S3 BUCKET ENUMERATION

Discover and identify exposed S3 buckets on the internet using industry standard tools and techniques.



FOCUS
Reconnaissance
& Discovery

LAB 02



S3 BUCKET SECURITY TESTING / EXPLOITATION

Assess S3 bucket access controls, identify misconfigurations and extract data in an authorized lab environment.



FOCUS
Access Control
& Data Exposure

LAB 03



ASSESSING IAM PRIVILEGE-ESCALATION RISKS

Analyze IAM policies and roles to identify misconfigurations that could lead to privilege escalation.



FOCUS
Identity Security
& Permissions

LAB 04



DOCKER IMAGE VULNERABILITY ASSESSMENT

Scan Docker images using Trivy to identify vulnerabilities, misconfigurations and associated risks.



FOCUS
Container Security
& Vulnerabilities

TOOLS & TECHNOLOGIES



AWS



AWS CLI



S3



IAM



KMS



Docker



Trivy



Linux

WHAT YOU WILL GAIN

- Deep understanding of AWS cloud security
- Hands-on experience with real-world tools
- Ability to identify and assess security risks
- Stronger practical cybersecurity skills



VIDIT SHRINGI



KRITI PUROHIT



EDITION 2026



LINKEDIN

<https://www.linkedin.com/company/internshield.in>

MODULE :

CLOUD COMPUTING THREATS AND COUNTERMEASURES

LAB OVERVIEW • TOOLS • ENVIRONMENT • LEARNING SCOPE

AUTHORIZED TRAINING SCOPE

All cloud-security assessment activities described in this overview must be performed only against resources explicitly authorized for training. Do not test third-party, public, or production systems.

Scenario

Cloud computing is an emerging technology that delivers computing services such as online business applications, online data storage, and webmail over the Internet. Cloud implementation enables a distributed workforce, reduces organization expenses, and can support data security. As enterprises increasingly adopt cloud services, cloud systems have emerged as targets for attackers seeking unauthorized access to valuable data.

Therefore, it is essential to regularly perform authorized security testing on cloud systems to monitor their security posture. Cloud systems can also introduce availability concerns because shared services may expose a cloud-based service endpoint to concentrated flooding activity. Adequate security is vital because cloud-computing services are based on shared infrastructure and services.

LEGAL & ETHICAL REQUIREMENT

Cloud security testing may be illegal when performed without authorization. Follow organizational policies, applicable laws, and the defined laboratory scope. This course is intended for controlled, authorized training.

Objective

The objective of the lab is to study cloud-platform security assessment activities that include, but are not limited to:

- Performing S3 bucket enumeration
- Assessing misconfigured S3 buckets in an authorized training environment
- Analyzing privilege-escalation risk caused by IAM policy misconfiguration
- Performing vulnerability assessment on Docker images

Overview of Cloud Computing

Cloud computing refers to on-demand delivery of IT capabilities in which IT infrastructure and applications are provided to subscribers as metered services over a network. Cloud services are commonly classified into Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).

Service Model	Description	Security Emphasis
IaaS	Infrastructure resources delivered as a service	Identity, network, compute and configuration
PaaS	Managed platform for application development	Application, identity, dependency and configuration
SaaS	Ready-to-use cloud applications	Identity, access, data and tenant security

Lab Scope

Lab	Activity	Primary Tool / Platform	What Students Learn
01	S3 bucket enumeration	Lazys3 / authorized S3 environment	Cloud-storage exposure discovery
02	S3 bucket security assessment	AWS CLI / S3	Storage permissions and misconfiguration risk
03	IAM privilege-escalation assessment	AWS IAM	Excessive permissions and least privilege
04	Docker image vulnerability assessment	Trivy / Docker	Image vulnerabilities and remediation priorities

Required Tools & Environment

The following tools and technologies are required or referenced by the supplied lab outline. Install and configure them only in the designated training environment.

Tool / Platform	Purpose	Setup Requirement
Amazon Web Services (AWS)	Cloud platform used for S3 and IAM labs	Authorized training AWS account
Amazon S3	Object-storage service used for storage-security assessment	Training bucket/resources
AWS CLI	Command-line interface for authorized AWS assessment tasks	Configured on the lab machine
Lazys3	S3 bucket enumeration utility referenced by the source	Available in the approved security-testing environment
AWS IAM	Identity and access-management service	Training IAM identities/policies
Docker	Container image platform	Local training environment
Trivy	Container image vulnerability scanner	Installed in the security lab
Security Testing VM	Controlled workstation for lab activities	Recommended isolated VM
Internet / Network Access	Required for cloud-console connectivity	Use the designated lab network

SETUP PRINCIPLE

Prepare the environment before class: verify access, confirm the target resources are intentionally created for training, and ensure students do not use personal or production credentials.

Recommended Lab Environment

Component	Recommended Configuration
Workstation	Windows/Linux host capable of running a security-testing VM
Security VM	Parrot Security or equivalent authorized training VM
Cloud Account	Dedicated AWS training account or isolated classroom environment
IAM	Dedicated training identities with intentionally limited permissions
S3	Dedicated training buckets containing non-sensitive sample data
Docker	Local container runtime with approved training images
Trivy	Current compatible version in the training environment

Visual & Evidence Policy

This document is an informational lab overview rather than a step-by-step practical worksheet. No operational screenshots are included here. The practical lab document should contain screenshot placeholders only where students are expected to capture authorized evidence.

EVIDENCE HANDLING

Never place real passwords, access keys, private keys, tokens, production data, or other sensitive information into screenshots, reports, repositories, or classroom submissions.

Learning Outcomes

- Explain why cloud environments require continuous security assessment.
- Describe the security risks associated with exposed cloud storage.
- Recognize the relationship between IAM permissions and privilege-escalation risk.
- Understand the role of vulnerability assessment in container security.
- Identify the purpose of Lazys3, AWS CLI, IAM, Docker, and Trivy within the course.
- Document security findings and recommend appropriate countermeasures.
- Perform all activities within a defined and authorized training scope.

Course Delivery Sequence

Stage	Topic	Learning Focus
01	Cloud Security Context	Understand the threat landscape and the purpose of authorized assessment.
02	S3 Enumeration	Identify cloud-storage exposure within the approved training scope.
03	S3 Security Assessment	Analyze permissions and misconfiguration risk.
04	IAM Security	Study excessive permissions and privilege-escalation risk.
05	Container Security	Assess Docker images for known vulnerabilities.
06	Analysis & Countermeasures	Document findings, prioritize risk, and identify remediation.

Security Principles Applied Across All Labs

Authorization — Test only resources that are explicitly within the approved scope.

Least Privilege — Give identities and resources only the permissions required for the intended task.

Data Minimization — Collect only the evidence required to demonstrate a security finding.

Isolation — Use dedicated training accounts, buckets, identities, images and networks.

Evidence Protection — Redact credentials and sensitive information before saving evidence.

Remediation — A security assessment should lead to corrective controls and verification.

Lab Readiness Checklist

- ☐ Authorized AWS training account available
- ☐ S3 training resources prepared
- ☐ AWS CLI installed and configured for the lab
- ☐ Lazys3 available in the authorized security VM
- ☐ IAM training identities and policies prepared

- ☐ Docker installed
- ☐ Trivy installed and tested
- ☐ Sample/non-sensitive data prepared
- ☐ Students briefed on legal and ethical scope
- ☐ Evidence-redaction process explained

INSTRUCTOR PREPARATION

Create intentionally vulnerable resources before the practical sessions and restore or delete them after use. The supplied source identifies four learning activities; this overview intentionally does not include attack commands or operational exploitation steps.